

Motivation

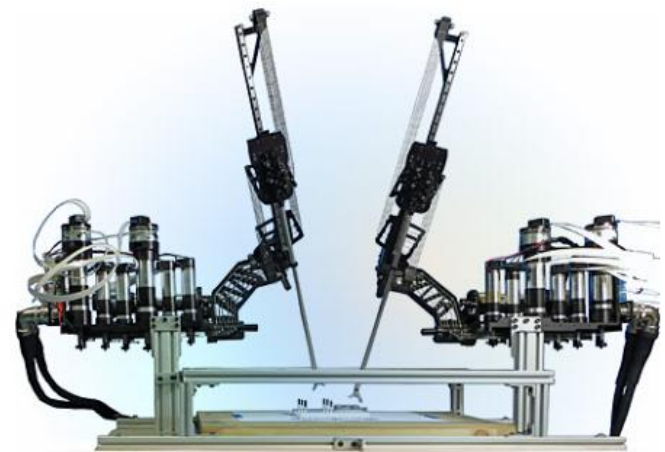


Figure 1: RAVEN cable-actuated surgical robot [1]. Manipulators must be replaced frequently for safety.

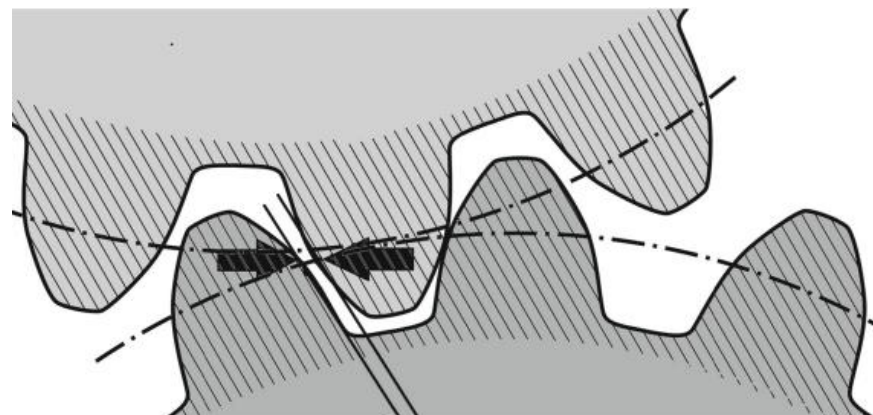


Figure 2: Visual of backlash in traditional gear transmission [2]. Cables avoid these unideal effects.

- Cables offer **extremely low backlash and inertia**.
- **However**, most cable-driven manipulators are only rated for **25 uses** due to cable stretch and creep concerns [3]

Theory

Cable-Actuated Reinforcement Learning algorithm (CARL) to robustly account for **tension variation** and **cable stretch**.

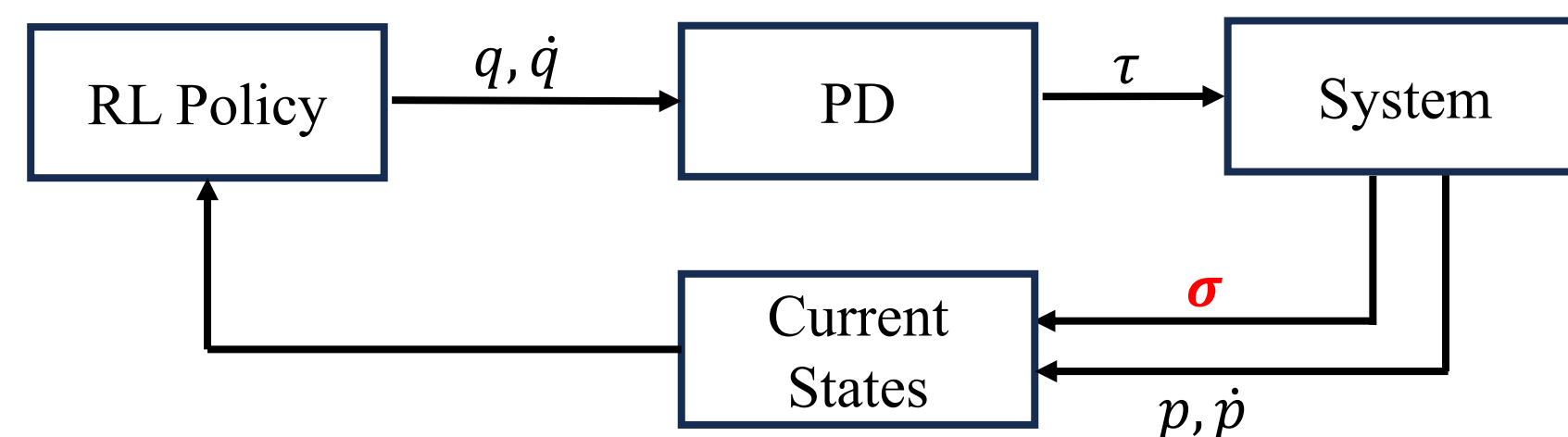


Figure 3: Control architecture for proposed approach. RL sends joint commands that are tracked with a PD controller. Cable tension (σ) and end effector position are fed back into the policy.

Proximal Policy Optimization (PPO) [4]

RL algorithm that encourages efficient policy updates while remaining stable.

$$L^{CLIP} = \mathbb{E}[\min(r_t A_t, \text{clip}(r_t, 1 - \epsilon, 1 + \epsilon) A_t)] \quad (1)$$

$$r_t = \frac{\pi(a_t, s_t)}{\pi_{\text{old}}(a_t, s_t)} \quad (2)$$

Cable representations

- Cables are modeled using **spring tendon** in MuJoCo simulation environment
- Tendons allow cable transmission without extra joints, keeping the **actuation realistic but the model simple**

Mechanical Design

Link	Weight [kg]	KUKA KR6 Weight [kg] [5]	% Reduction
Elbow	1.4 ± 0.1	14 ± 2	$90 \pm 2\%$
Wrist	0.7 ± 0.1	8 ± 1	$91 \pm 2\%$

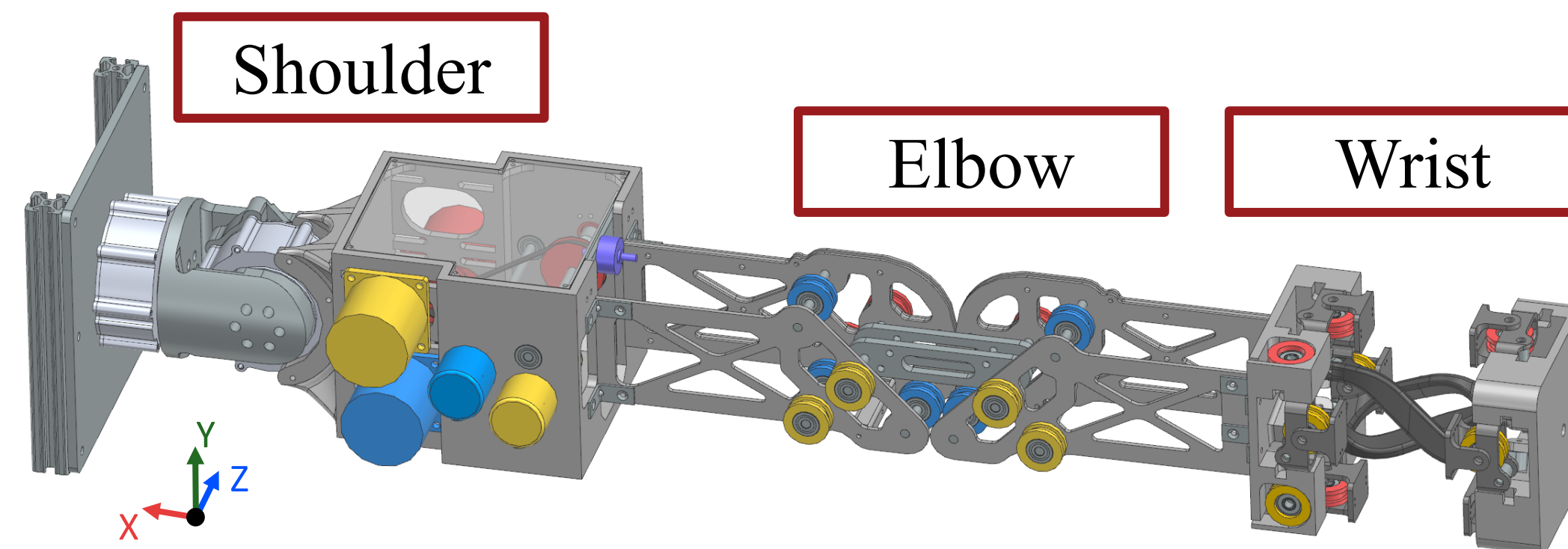


Figure 4: Full CAD model inspired by [6]. Arm span from shoulder to wrist is $32.75 \pm 0.06\text{in}$, which is comparable to the average male arm length of 30.8in. Motors, pulleys, and encoders of each cable path are color coded.

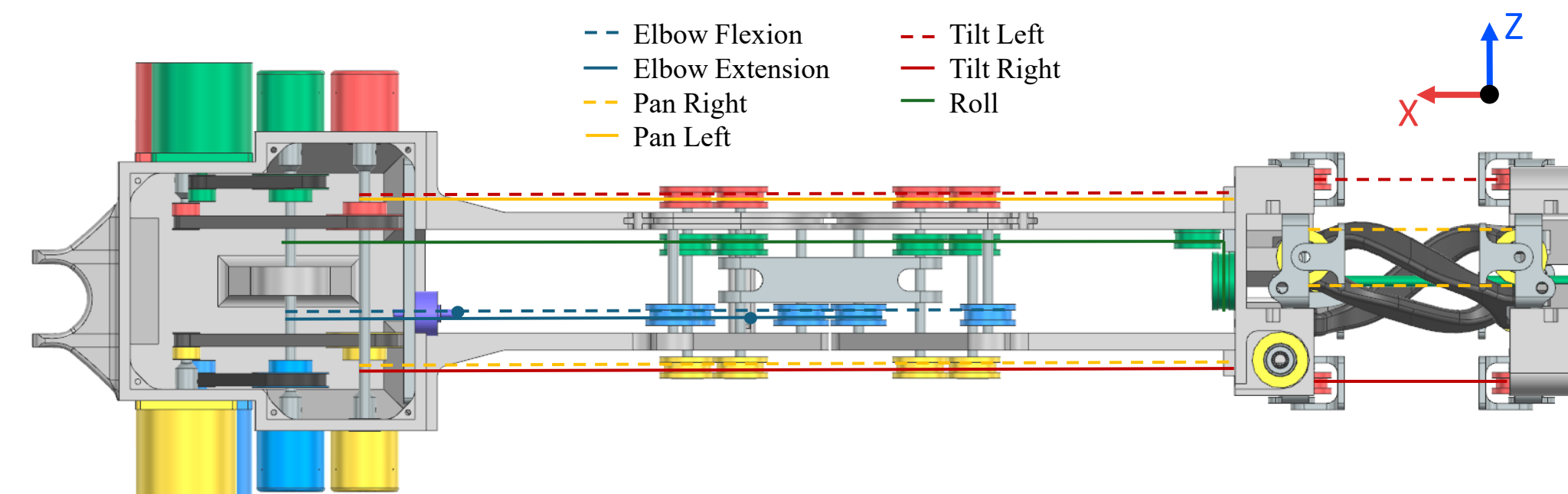


Figure 5: Full cable routing schematic. Elbow cable is terminated on the load cell (purple) for tension feedback.

Experimental Design

- Positional accuracy tested with 0.001" precision indicator

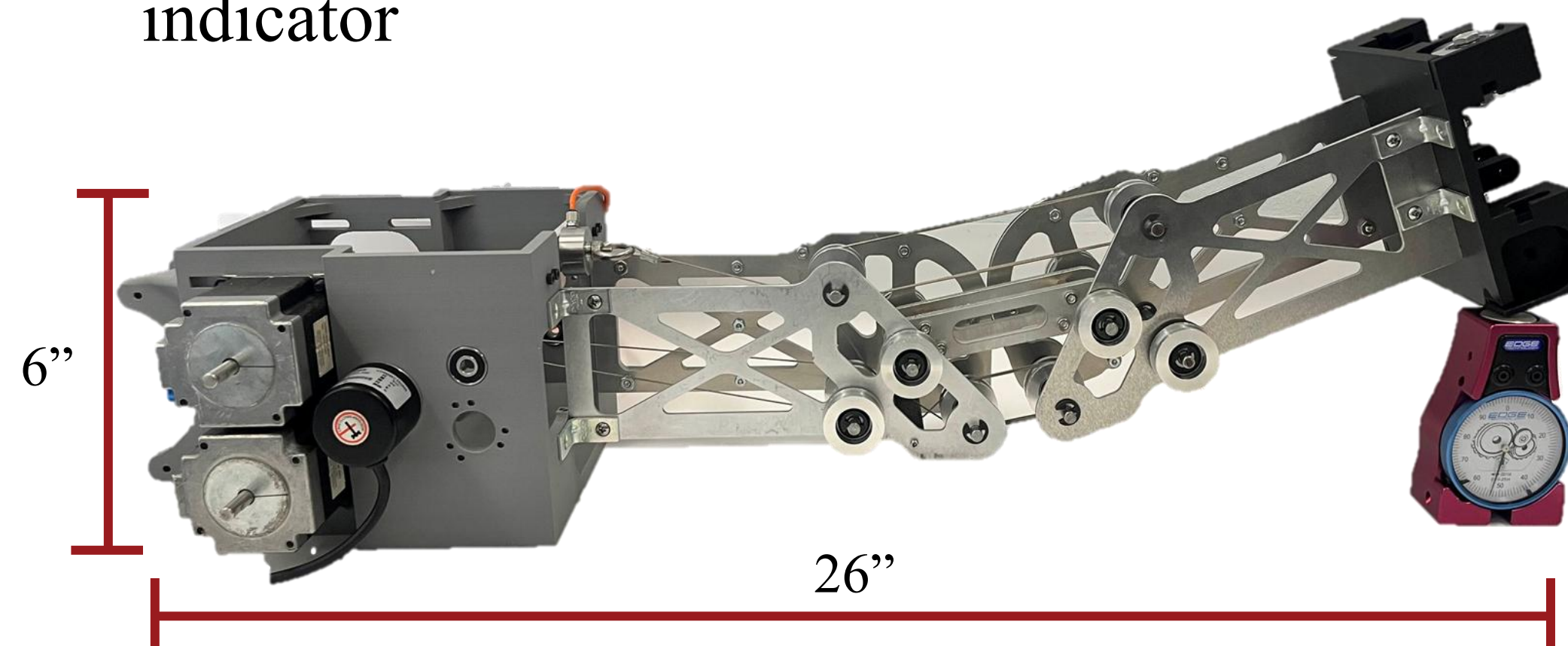


Figure 6: Representative photo of elbow actuating onto machining indicator to test positional repeatability.

Results

Simulation vs Commanded Trajectory

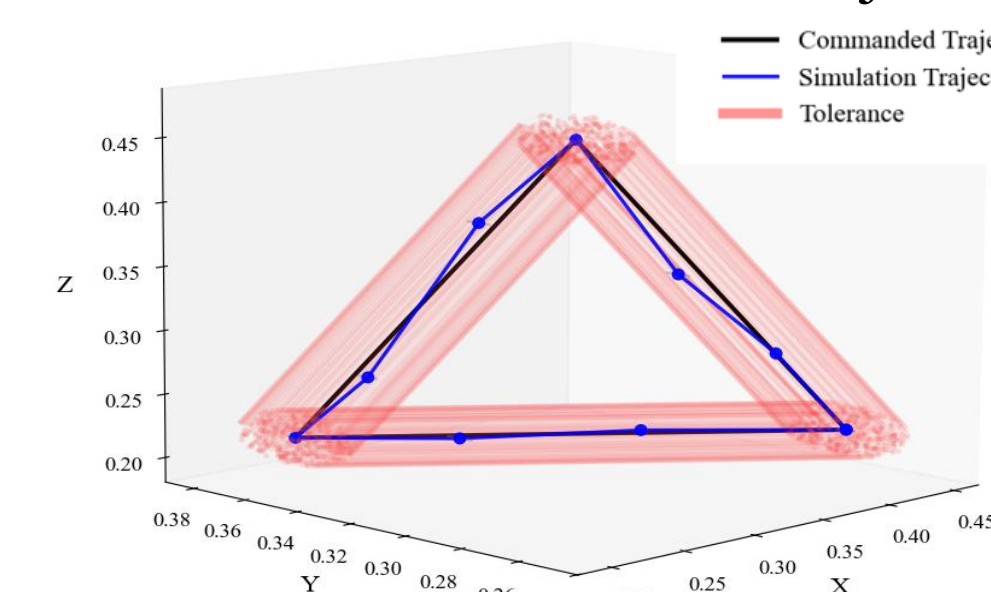


Figure 7: Plot of commanded and simulated trajectory. Tolerance of 4 cm is given when training the model.

Real vs Sim Tension

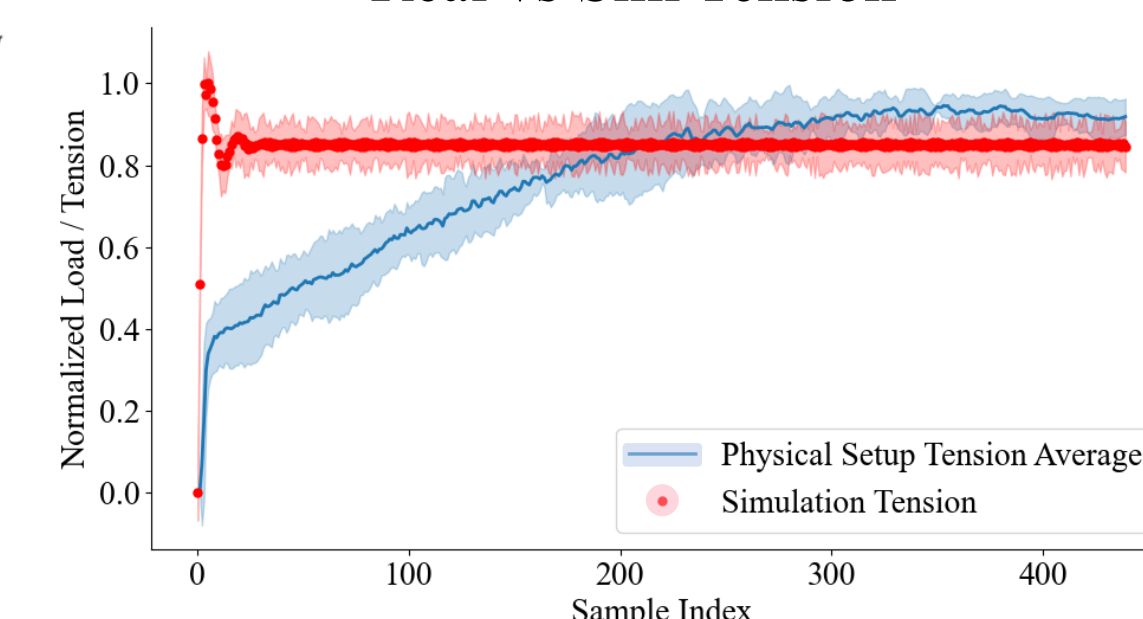


Figure 8: Simulation modeling of tendons captures cable tension at the end of an action, but not the beginning due to cable winding/nonlinear effects.

Tension Magnitudes With and Without Tension Awareness

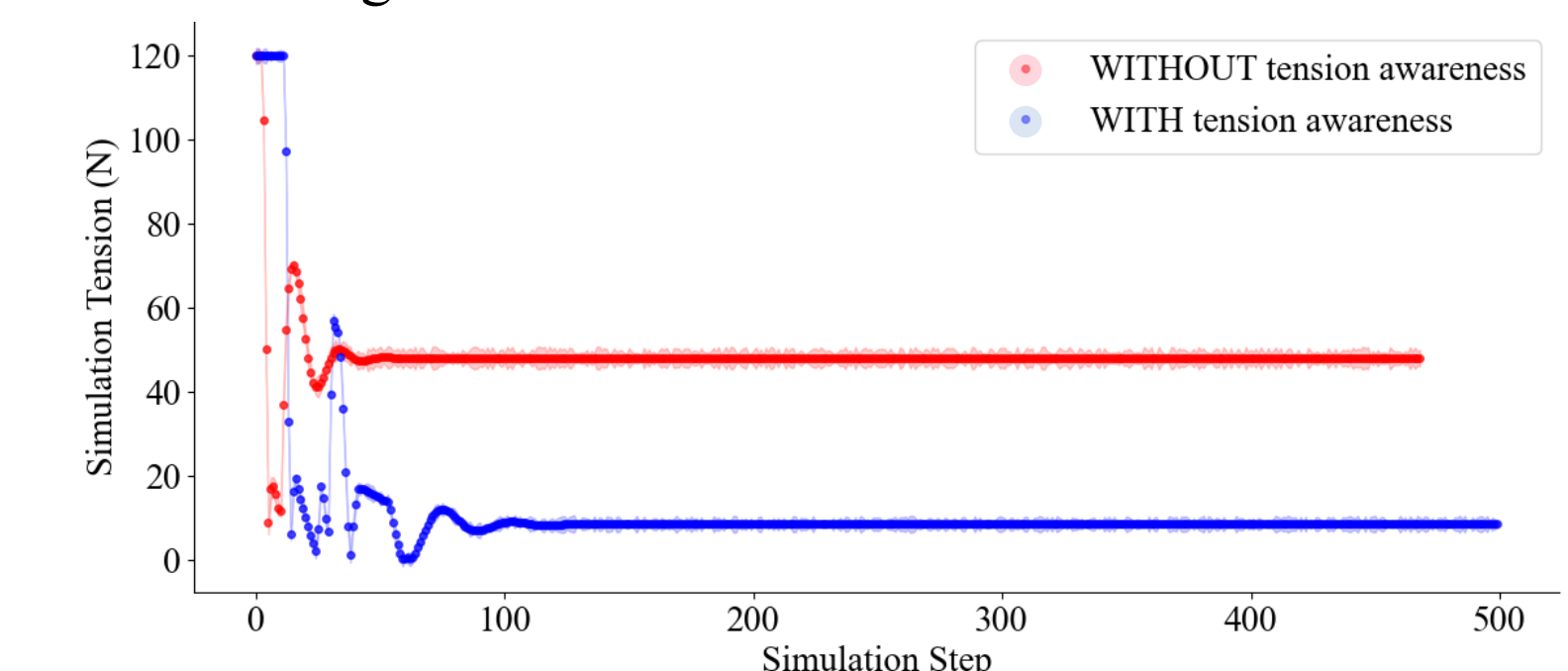


Figure 9: After settling, tension-aware RL maintains a steadier and lower tension magnitude.

Conclusion

- CARL can lift **5kg** and has repeatable precision of $\pm 0.004\text{in}$ for fine movements
- RL successfully deployed but requires tuning
- Cost : **\$1350** (~80% is from motors)

Future Work

- Sim-to-real tuning
- Inclusion of wrist actuation with tension awareness
- Further investigation of cable representations

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References

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